



## Department of Electronics and Communication Engineering

Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: B.E., III Semester & CSE

Course Code & Title: EC8395 & Communication Engineering

Name of the Faculty member (s): Dr.S.Vairaprakash

### Innovative Practice Description

- **Unit / Topic:** V/Multiple Access Techniques
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO15
- **Activity Chosen:** Flipped Class room
- **Justification:**

The topic was chosen for flipped class because Multiple Access techniques covers TDMA, FDMA, CDMA. The Unit 5 deals Direct Sequence Spread Spectrum and Multiple Access Techniques. Since the students are exposed to the knowledge of mobile operations, the topic of “Multiple Access Techniques” for flipped class was given. This topic can be learnt in self-learning mode with simple logic. The students have chance to browse the topic to know how the accessing techniques are using in satellite and mobile communications. And find more information related to error detection and correction.. Thus this flipped class activity is chosen to make them to work out different techniques for mobile and satellite.

- **Time Allotted for the Activity:** 45 minutes
- **Details of the Implementation:**

The overview of flipped class activity was given to students before the implementation of the activity. The students were divided into groups and one topic is allotted to each group from Multiple Access Techniques topic as their wish. The topic related information and e-sources were given to the students. The students have to prepare a presentation for 10 minutes and each group has to present their topic chosen and others will listen to the presentation. After the presentation, question & answering is planned and others can share their ideas and doubts. By this way, all groups were shared their concepts learnt in their topic inside the class.

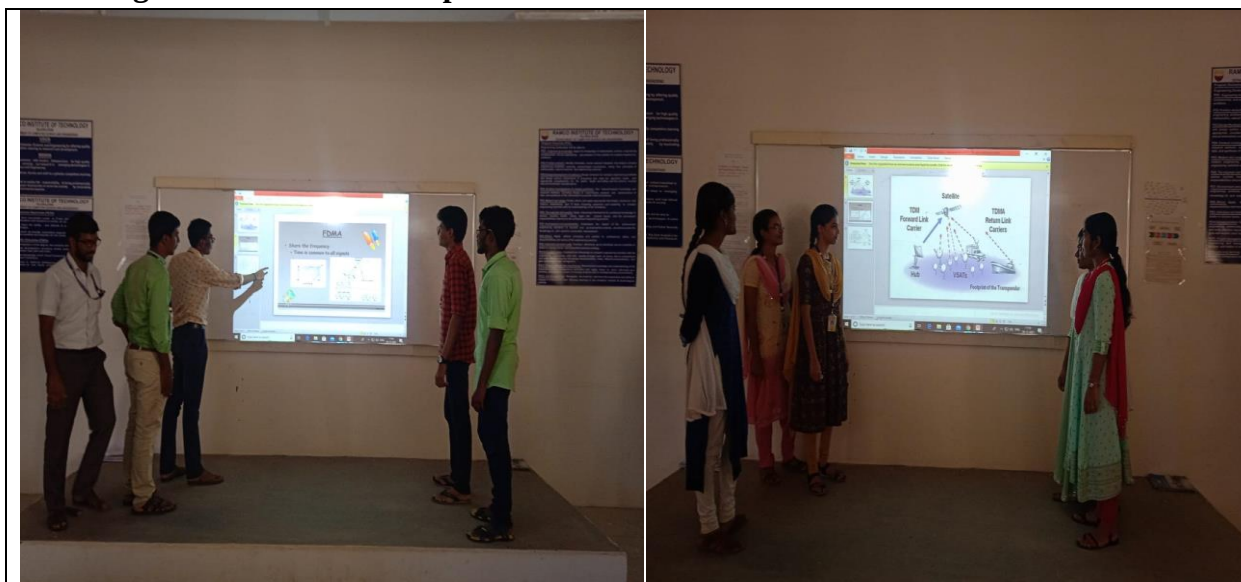
- **CO – PO / PSO mapping:**

| CO  | PO1 | PO2 | PO3 | PO4 | PO10 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|------|------|------|------|
| CO5 | 3   | 2   | 2   | 2   | 1    | 1    | 3    | 2    |

- **PO / PSO mapped:**

| Innovative practice                  | PO4                                                                                                                                                                                  | PSO1                                                                                                                                   |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
|                                      | 2                                                                                                                                                                                    | 3                                                                                                                                      |
| <b>Justification for correlation</b> | The students will be able to study the relationship between different access Techniques and underlying principles of accessing the frequency spectrum hence it is mapped to level 2. | The students will be able to apply the knowledge of multiple access techniques in mobile communication; hence it is mapped to level 3. |

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

The students felt a new practice and it helped them to know to make good presentation and avoid stage fear. They liked this practice of knowledge sharing that can be done through this type of activity. They could get idea from learning sources which would help them to prepare the presentation.

- ❖ **Benefit of the practice:**

Students done this activity in a successful manner. They planned the activity for their chosen topic and prepared the presentation in the conceptual manner. All members in the group were learnt the topics thoroughly. By forming the group, students were able to grasp the knowledge of concept by sharing the ideas among them. Each member in the group was able to present and deliver the concept of different multiple access techniques and underlying principles of accessing the frequency spectrum

❖ ***Challenges faced in implementation:***

Some of the group students prepared minimal content for presentation

Some of the students felt difficult to present because of lack of communication.

Few groups done their presentation beyond their time limit.

Few students in the group felt difficult to answer all the questions.

**References:**

1. B.Sklar, “Digital Communication Fundamentals and Applications”, 2nd Edition, Pearson Education, 2009.
2. B.P.Lathi, “Modern Digital and Analog Communication Systems” 3rd Edition, Oxford University Press 2007.
3. H P Hsu, Schaum Outline Series - —Analog and Digital Communications, TMH 2006.
4. J.G.Proakis, “Digital Communication”, 4th Edition, Tata McGraw Hill Company, 2001.
5. <https://www.panopto.com/blog/what-is-a-flipped-classroom/>



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Course Code & Title: EC8395& Communication Engineering

Name of the Faculty member (s): Dr.S.Vairaprakash

### Innovative Practice Description

- **Unit / Topic:** III/ BER comparison of Digital Modulation Schemes
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO8
- **Activity Chosen:** Jigsaw
- **Justification:**

The activity “JIGSAW” is used to recollect the concepts knowledgeable and to enrich the learning through this teamwork. The concept of generation, detection and error probability of various digital modulation schemes were already learnt in modulation techniques. By recollecting all the information in various modulation schemes, students from expert group were able to share the information to home group.

- **Time Allotted for the Activity:** 45 minutes
- **Details of the Implementation:**

The students are divided into Home Groups with each group having 6 members . Each member in home group will form different groups named Expert group. Each expert group is allotted with each modulation technique The expert group members discuss the topic in detail for 20 minutes with the learning materials and e-sources

The Expert group members should go to their original home group and discuss the points (25 Minutes) to the other members. All the members in home group learnt about all the topics through expert group members.

#### CO – PO / PSO mapping:

| CO  | PO1 | PO2 | PO3 | PO4 | PO5 | PO9 | PSO1 |
|-----|-----|-----|-----|-----|-----|-----|------|
| CO5 | 3   | 3   | 2   | 2   | 2   | 1   | 3    |

- **PO / PSO mapped:**

| Innovative practice                  | PO4                                                                                                                                                                        | PSO1                                                                                                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | 3                                                                                                                                                                          | 3                                                                                                                                                                                                   |
| <b>Justification for correlation</b> | The outcome is mapped to level 3 because digital Modulation schemes are designed ad experiments are conducted to evaluate the probability of error occurs in transmission. | The students will be able to understand the band pass transmission schemes for designing the special antennas and the underlying circuitry in the area of communication, hence it is highly mapped. |

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

The students felt a new experience and they could able to recollect the information about digital modulation techniques from their class mates. It helped them to learn the concept through collaborative learning with their class mates. They liked this practice of learning through this type of activity for recollecting the information that they already learnt.

- ❖ **Benefit of the practice:**

By forming the group, students were able to grasp the knowledge of concept by sharing the ideas among them. Each Home group could recollect the information of seven modulation techniques .

- ❖ **Challenges faced in implementation:**

Some of the students faced difficulty to elaborate the concepts.  
Some students share minimal information to their home group.

## References:

1. B.P.Lathi, —Modern Digital and Analog Communication Systems, 3rd edition, Oxford University Press, 2007
2. H P Hsu, Schaum Outline Series – —Analog and Digital Communications, TMH 2006 3.B.Sklar, Digital Communications Fundamentals and Applications, 2/e Pearson Education 2007.
- 3.J.G.Proakis, “Digital Communication”, 4th Edition, Tata McGraw Hill Company, 2001.



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Department of Electronics and Communication Engineering

Academic Year: 2021- 2022 (Odd Semester)

## Innovative Teaching Method

### UNIT I – Analog Modulation

Degree, Semester & Branch: III Semester B.E. CSE

Course Code & Title: EC8395 Communication Engineering

Name of the Faculty member: Dr.S.Vairaprakash

Name of the Topic: AM Generation and Demodulation

Name of the Innovative Practice: One-minute Paper

Date & Duration: 24.08.2021 & 5 minutes

Sample Images:

| Innovative Teaching Method Execution                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AM Generation and Demodulation                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                           |
| <p>953620104008<br/>II CSE</p> <p>AM equation<br/> <math display="block">V_{AM} = E_c [1 + m_a \sin \omega_m t]</math>           where <math>m_a = \frac{E_m}{E_c} \Rightarrow</math> Modulation Index</p> <p>Bandwidth = <math>2f_m</math></p> <p>Spectrum</p> | <p>953620104303<br/>II CSE</p> <p>Power equation <math>P_t = P_c [1 + \frac{m_a^2}{2}]</math></p> <p>In DSBSC <math>\Rightarrow</math> Power saving = 66.7%.</p> $m_a = \frac{E_{max} - E_{min}}{E_{max} + E_{min}}$ <p>Bandwidth = <math>2f_m</math></p> $V_{AM}(t) = E_c \sin \omega_c t + \frac{m_a E_c}{2} \cos(\omega_c - \omega_m)t + \frac{m_a E_c}{2} \cos(\omega_c + \omega_m)t$ |



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## **Innovative Teaching Method**

### **UNIT I – Analog Modulation**

**Degree, Semester& Branch: III Semester B.E. CSE**

**Course Code & Title: EC8395 Communication Engineering**

**Name of the Faculty member: Dr.S.Vairaprakash**

**Name of the Topic: Super heterodyne Receiver**

**Name of the Innovative Practice: Quiz**

**Date & Duration: 6.09.2021 & 15 minutes**

### **Innovative Teaching Method Execution**

#### **Super heterodyne Receiver**





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## Innovative Teaching Method

### UNIT I – Analog Modulation

Degree, Semester& Branch: III Semester B.E. CSE

Course Code & Title: EC8395 Communication Engineering

Name of the Faculty member: Dr.S.Vairaprakash

Name of the Topic: FM Modulators and Demodulators

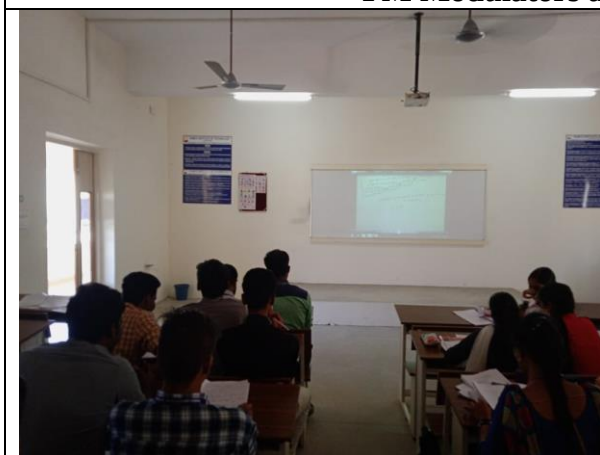
Name of the Innovative Practice: NPTEL Video

Date & Duration: 07.09.2021 & 10 minutes

Sample Images:

#### Innovative Teaching Method Execution

#### FM Modulators and Demodulators





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## Innovative Teaching Method

### UNIT II –Pulse Modulation

Degree, Semester& Branch: III Semester B.E. CSE

Course Code & Title: EC8395 Communication Engineering

Name of the Faculty member: Dr.S.Vairaprakash

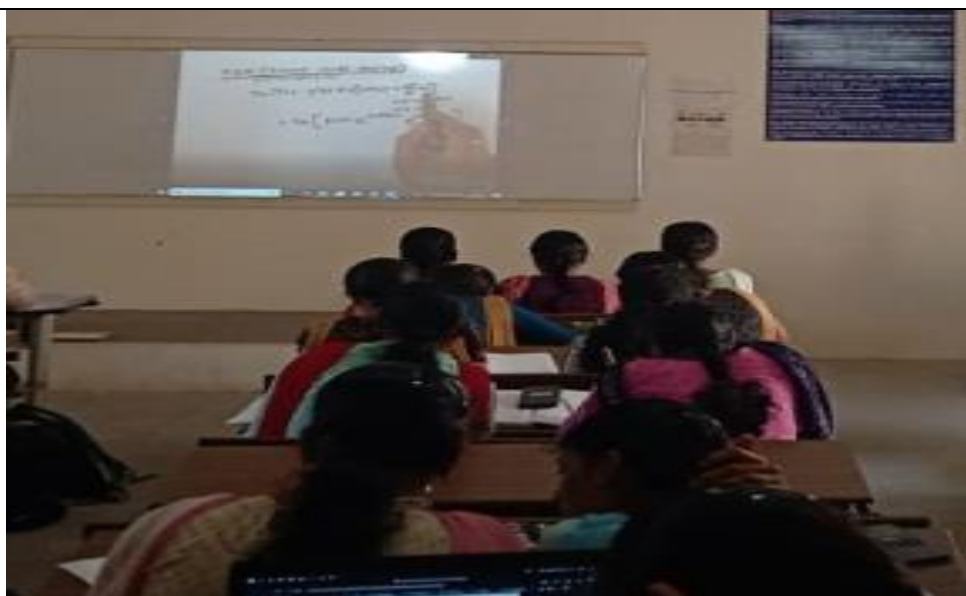
Name of the Topic: Line codes

Name of the Innovative Practice: NPTEL Video

Date & Duration: 20.09.2021 & 5 minutes

## Innovative Teaching Method Execution

### Line codes





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## **Innovative Teaching Method**

### **UNIT II –Pulse Modulation**

**Degree, Semester& Branch: III Semester B.E. CSE**

**Course Code & Title: EC8395 Communication Engineering**

**Name of the Faculty member: Dr.S.Vairaprakash**

**Name of the Topic: ADPCM&ADM Principles**

**Name of the Innovative Practice: Role Play**

**Date & Duration: 27.09.2021 & 5 minutes**

## **Innovative Teaching Method Execution**

### **ADPCM&ADM Principles**





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## **Innovative Teaching Method**

### **UNIT II –Pulse Modulation**

**Degree, Semester& Branch: III Semester B.E. CSE**

**Course Code & Title: EC8395 Communication Engineering**

**Name of the Faculty member: Dr.S.Vairaprakash**

**Name of the Topic: Frequency Division Multiplexing**

**Name of the Innovative Practice: Zero Minute Speech**

**Date & Duration: 05.10.2021 & 5 minutes**

## **Innovative Teaching Method Execution**

### **Frequency Division Multiplexing**





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## Innovative Teaching Method

### UNIT III – Digital Modulation and Transmission

Degree, Semester & Branch: III Semester B.E. CSE

Course Code & Title: EC8395 Communication Engineering

Name of the Faculty member: Dr.S.Vairaprakash

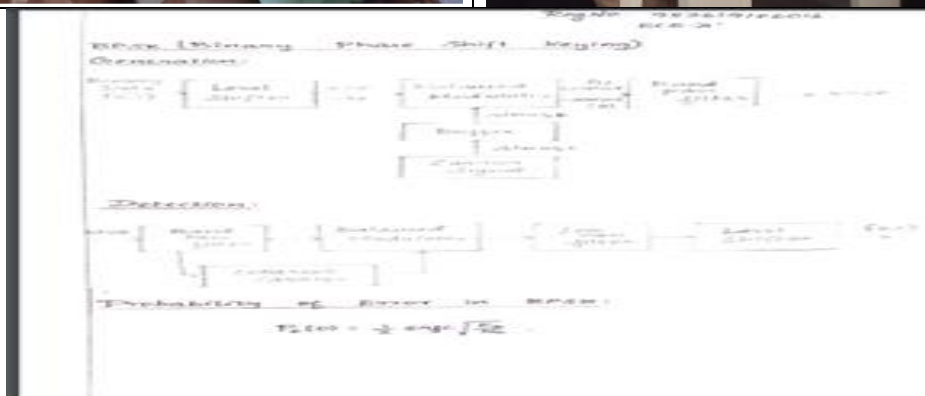
Name of the Topic: Generation, Detection, PSD and BER of coherent BPSK

Name of the Innovative Practice: Sketch Noting

Date & Duration: **06.10.2021 & 5 minutes**

#### Innovative Teaching Method Execution

#### Generation, Detection, PSD & BER of Coherent BPSK





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## **Innovative Teaching Method**

### **UNIT III – Digital Modulation and Transmission**

**Degree, Semester & Branch: III Semester B.E. CSE**

**Course Code & Title: EC8395 Communication Engineering**

**Name of the Faculty member: Dr.S.Vairaprakash**

**Name of the Topic: Generation, Detection of coherent QPSK and DPSK**

**Name of the Innovative Practice: Think Pair Share**

**Date & Duration: 11.10.2021 & 45 minutes**

### **Innovative Teaching Method Execution**

#### **Generation, Detection of coherent QPSK and DPSK**





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## **Innovative Teaching Method**

### **UNIT III – Digital Modulation and Transmission**

**Degree, Semester& Branch: III Semester B.E. CSE**

**Course Code & Title: EC8395 Communication Engineering**

**Name of the Faculty member: Dr.S.Vairaprakash**

**Name of the Topic: Principles of M-ary PSK**

**Name of the Innovative Practice: NPTEL Video**

**Date & Duration: 12.10.2021 & 45 minutes**

### **Innovative Teaching Method Execution**

#### **Principles of M-ary PSK**





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## **Innovative Teaching Method**

### **UNIT IV –Information Theory and Coding**

**Degree, Semester& Branch: III Semester B.E. CSE**

**Course Code & Title: EC8395 Communication Engineering**

**Name of the Faculty member: Dr.S.Vairaprakash**

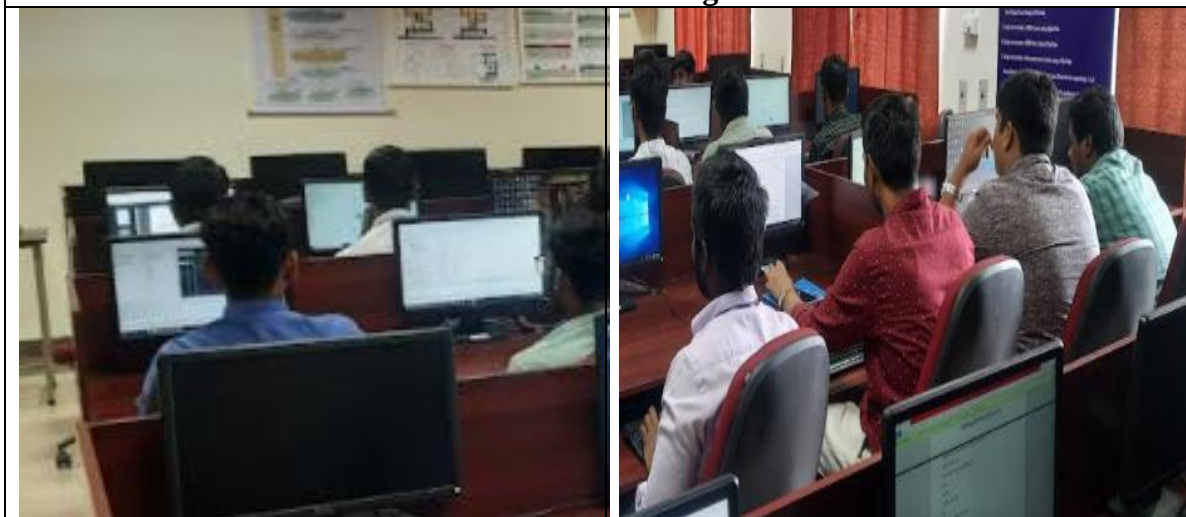
**Name of the Topic: Huffman Coding**

**Name of the Innovative Practice: Simulation**

**Date & Duration: 12.11.2021 & 10 minutes**

### **Innovative Teaching Method Execution**

#### **Huffman Coding**





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## **Innovative Teaching Method**

### **UNIT IV –Information Theory and Coding**

**Degree, Semester& Branch: III Semester B.E. CSE**

**Course Code & Title: EC8395 Communication Engineering**

**Name of the Faculty member: Dr.S.Vairaprakash**

**Name of the Topic: Syndrome Decoder**

**Name of the Innovative Practice: Roleplay**

**Date & Duration: 12.11.2021 & 10 minutes**

### **Innovative Teaching Method Execution**

#### **Syndrome Decoder**

